Assignment 6C. Gaussian quadrature Marks 20

Posted on 17.09.2026 @ 2:30 pm and due on 17.09.2026 @ 5:40 pm

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1. Write library functions for Gauss-Legendre and Gauss-Laguerre quadrature for numerical integration. Hardwire the table for zeros and weights for Legendre and Laguerre to any order of your choice (at least $n = 5$). There should be only one Gauss Quadrature routine and depending on what you want it should choose appropriate table for Legendre and Laguerre. **[5]**

2. Use appropriate $n = 4$-point Gaussian quadrature to evaluate

$$\int_{-1}^{1} \frac{x^2}{1+x^4}\,dx \;\approx\; 0.487\,495\,494$$

   Compare this with Simpson's 1/3-rule up to the same precision. **[5]**

3. Use appropriate Gaussian quadrature to evaluate the following integral with $n = 5$. **[5]**

$$\int_0^{\infty} \frac{e^{-x}}{1+x}\,dx, \qquad \text{analytical value} \;\approx\; 0.59635$$

4. Integrate the following with appropriate n-point Gauss-Laguerre

$$\int_0^{\infty} e^{-x}\,x^4\,dx$$

   Compare with the analytical result $\int_0^{\infty} \exp(-x)\,x^n = n!$ **[5]**